Reflection on Code

Credit Name: CSE2130 - File Structures & Exception Handling
Assignment Name: Student Semester GUI

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem
How did you approach understanding the challenge?
Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I approached the challenge by figuring out how to make a program that could calculate a student's semester average and save their information to a text file. I had to understand how to take information from text fields and use it in calculations. I worked through this by looking at examples and testing different parts of my code.

Planning the Solution
Did you create a plan or break the problem into smaller steps before coding?
How did you decide on the tools, data structures, or algorithms to use?

I broke the program into smaller steps. First, I created the GUI with text fields for the student's information and grades. Then, I added the calculation for the average, followed by the code to save the information to a file and display the saved records. I used JTextField for input, JLabel to display the average, and file-handling classes to read and write student records.

Implementation
Did you write the code in small pieces or attempt the entire solution at once?
How did you handle moments when you felt stuck or unsure of what to do next?

I built the program in smaller pieces, starting with the GUI and then adding the buttons and their actions. I tested the different features as I worked on them to make sure they functioned properly. When I was unsure how to calculate and display the percentage or make the text file work, I used examples and AI to help me understand the code.

Overcoming Challenges
What part of the problem was the most difficult for you?
How did you handle moments when you felt stuck or unsure of what to do next?

The most difficult part was getting the program to calculate the average correctly and save the information to a file. I also had to make sure the program could read the saved information and display it in the text area. I worked through these challenges by breaking the code into smaller sections and using error handling to deal with invalid grade inputs and file errors.

Learning
Was there anything you learned that you think will help you with future challenges?

I learned how to use file handling to save and read information, calculate averages using multiple grades, and display results in a GUI. I also learned how to use try-catch statements to handle errors. These skills will help me with future programming assignments because I can use them to build programs that store information and handle user input more effectively.